

Inputs — FS Final-Drive Gearbox

Drivetrain Input (from CBR600RR primary-reduction output shaft)

Input speed, n_{in}	11500 rpm
Input torque, T_{in}	63.0 N·m

Ratio (fixed tooth counts, per spec)

Stage 1 ratio, i_1 ($z_1=20, z_2=40$)	1.58
Stage 2 ratio, i_2 ($z_1=24, z_2=48$)	1.58
Overall ratio achieved, $i_1 \times i_2$	2.50

Efficiency & Resulting Output

Assumed efficiency per stage	95.0%
Overall efficiency (2 stages)	90.3%
Output speed	4606.6 rpm
Output torque	141.9 N·m

Manufacturing & Mounting (confirmed)

ISO 1328 quality grade	6
AGMA-equivalent quality number, $Q_v = 17 -$	6.0
Application factor, K_A (as specified)	1.50
Design life target	20 hours

Gear Geometry Common Inputs

Normal pressure angle, α_n	20 deg
Helix angle, β (spur = 0)	0 deg

Material Gears -Case-Carburised Steel 16 Mn Cr 5

Bending endurance limit, σ_{Flim} (ISO ϵ	430 MPa
Contact endurance limit, σ_{Hlim} (ISO ϵ	1500 MPa

Material Shaft - 42CrMo4

Ultimate Tensile Strength S_{Ut}	900 MPa
Yield Strength S_y	680 MPa
Minimum Value	680 MPa